28/Aug/2020

1. $y'' + 9y = x(t)$, $x(t) = 8\sin t$ if $0 < t < \pi$ and 0 if $t > \pi$

$y(0) = 0$

$y'(0) = 4.$

$$L\{y''\} = s^2 L\{y\} - s\,y(0) - y'(0)$$

$$L\{y'\} = s\,L\{y\} - y(0).$$

$$L\{y''\} + 9L\{y\} = 8\,L\{\sin t\}$$

$$9L\{y\} + s^2 L\{y\} - s(0) - (4) = \frac{8}{s^2+1}$$

$$L\{y\}(9+s^2) - 4 = \frac{8}{s^2+1}$$

$$L\{y\} = \frac{8}{s^2+1} + 4$$

$$L\{y\} = \frac{8}{(s^2+1)(9+s^2)} + \frac{4}{9+s^2}$$

$$y = L^{-1}\left\{\frac{8}{(s^2+1)(9+s^2)}\right\} + 4\sin 3t.$$

$$\frac{1}{(s^2+1)(9+s^2)} = \frac{A}{s^2+1} + \frac{B}{9+s^2}$$

$$As^2 + A9 + Bs^2 + B$$

$$(A+B)s^2 + 9A + B = 1$$

$$8A = 1$$

$$A = 1/8$$

$$L^{-1}\left\{\frac{1}{s^2+1} - \frac{1}{s^2+9}\right\} + 4\sin 3t + B = -1/8$$

$$\sin t - \sin 3t + 4\sin 3t$$

$$3\sin 3t + \sin t$$

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2. $y'' + y = r(t) \quad r(t) = t$

$y(0) = y'(0) = 0$.

$L\{y''\} + L\{y\} = \frac{1}{s^2}$

$s^2 L\{y\} + L\{y\} = \frac{1}{s^2}$

$L\{y\}(s^2+1) = \frac{1}{s^2}$

$L\{y\} = \frac{1}{s^2(s^2+1)}$

$y = L^{-1}\left\{\frac{1}{s^2(s^2+1)}\right\}$

$\frac{1}{s^2(s^2+1)} = \frac{A}{s^2} + \frac{B}{s^2+1}$

$1 = A(s^2+1) + Bs^2$

$= (A+B)s^2 + A$

$A = 1 \quad B = -1$

$y = L^{-1}\left\{\frac{1}{s^2} - \frac{1}{s^2+1}\right\}$

$= t - \sin t$.

3. $y'' + y = \delta(t - 2\pi)$

$y(0) = 10 \quad y'(0) = 0$.

$(1+s^2)L\{y\} - 10 = L\{\delta(t-2\pi)\}$

$= \frac{e^{-2\pi s} + 10}{s^2+1}$

$y = L^{-1}\left\{\frac{e^{-2\pi s}}{s^2+1}\right\} + 10\sin t$.

$= u(t - 2\pi)\, L^{-1}\left\{\frac{1}{s^2+1}\right\}_{t \to t-a} + 10\sin t$

$u(t-2\pi) \sin(t-2\pi)$

4. $y'' + 2y' + 5y = 25t - 100\,\delta(t-\pi)$ $\quad y(0) = -2$, $y'(0) = 5$

$$L\{y''\} + 2L\{y'\} + 5L\{y\} = \frac{25}{s^2} - 100\,e^{-\pi s}$$

$$s^2 L\{y\} + 2s - 5 + 2[sL\{y\} + 2] + 5L\{y\} = \frac{25}{s^2} - 100\,e^{-\pi s}$$

$$s^2 L\{y\} + 2s - 5 + 2sL\{y\} + 4 + 5L\{y\}$$

$$L\{y\}(s^2 + 2s + 5) + 2s - 1 = \frac{25}{s^2} - 100\,e^{-\pi s}$$

$$L\{y\} = \frac{25}{s^2(s^2+2s+5)} - \frac{100\,e^{-\pi s}}{s^2+2s+5} - \frac{2s}{s^2+2s+5} + \frac{1}{s^2+2s+5}$$

$$L^{-1}\left\{\frac{25}{s^2(s^2+2s+5)}\right\} = L^{-1}\left\{\frac{1}{(s+1)^2+4}\right\} = e^{-t}\sin 2t.$$

$$\cancel{\frac{As+B}{s^2} + \frac{Cs+D}{s^2+2s+5}}$$

$$\cancel{As+B(s^2+2s+5) + Cs^2 + Ds^2}$$

$$\cancel{As^3}$$

$$\frac{A}{s} + \frac{B}{s^2} + \frac{Ds+E}{s^2+2s+5}.$$

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$$\frac{-2}{s} + \frac{5}{s^2} + \frac{2(s+1)-3}{(s+1)^2+4}$$

$$L^{-1} = 5t - 2 + e^{t}\left(2\cos 2t - \frac{3}{2}\sin 2t\right)$$

$$100L^{-1}\left[\frac{e^{-\pi s}}{s^2+2s+5}\right] = 50\,u(t-\pi)\,e^{-(t-\pi)}\sin 2t$$

$$L^{-1}\left[\frac{1}{s^2+2s+5}\right] = \frac{e^{-t}}{2}\sin 2t$$

$$L^{-1}\left[\frac{2s}{s^2+2s+5}\right] = e^{-t}\left[2\cos 2t - \sin 2t\right]$$

Total $\rightarrow$ $5t - 2 - 50\,u(t-\pi)\,e^{-t+\pi}\sin 2t$.

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